Design and Actuation of Conformal Magneto-Metamaterials

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Background and Motivation

Metamaterials are engineered materials with unique mechanical and wave manipulation properties enabling rapid shape changes. Magneto-mechanical metamaterials, controlled through magnetic fields, achieve property tuning via shape reconfiguration. Hard-magnetic soft materials, created by embedding hard magnetic particles into a soft polymeric matrix, undergo complex, untethered, reversible, and rapid deformations in response to magnetic fields.[1]

This research focuses on hard-magnetic soft materials, investigating 3D substrate structures such as concave, convex, and cylindrical shapes. We perform experimental tests and simulations to explore how these materials interact with electromagnetic fields. Additionally, we study the deformation under different amperages and concentrations of magnetic powder.

Previous research has shown acoustic 2D magneto-metamaterials can filter, regulate wave transmission, and steer acoustic waves, demonstrating the broader versatility of metamaterials.[2] Through manufacturing and designing 3D metamaterial substrates similar tests can be performed highlighting the importance of continued investigation into electromagnetic deformation.

3D Design

Unique cell molds were designed for each metamaterial section based on the specific 3D shape. While the height of the cells remained constant, the width and side angles varied to ensure proper conformation to the 3D surface. This customization was essential for the array to fit correctly and function effectively.

For example, the cylindrical structure used cells with uniform dimensions and connection points to maintain its shape. In contrast, the concave and convex structures required cells with adjusted widths and side angles to match their respective curvatures.

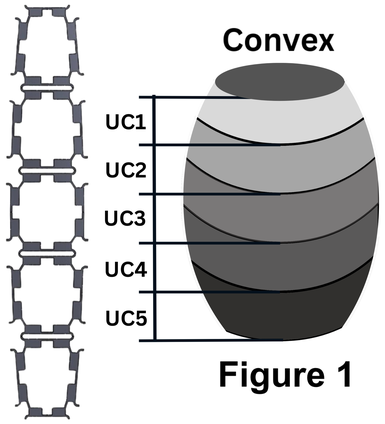
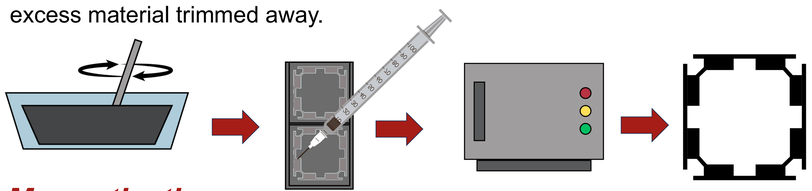


Figure 1

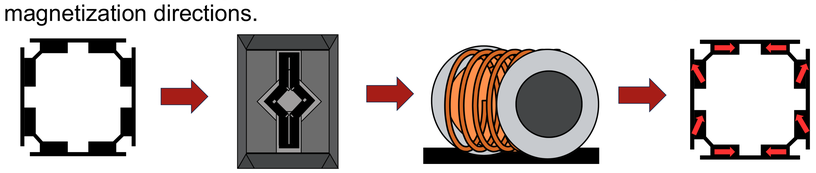
Fabrication

The existing metamaterial design and fabrication process developed by the lab was used to test all magneto-metamaterials. The unit cells were fabricated to ensure uniformity and functionality. We began by preparing a blend of DragonSkin 20 silicone elastomer and NdFeB magnetic particles. This mixture was degassed and then poured into 3D-printed molds made from PVA filament. The molds were heated at 80ºC for 30 minutes to cure the material. Post-curing, each unit cell was carefully extracted or dissolved from the molds, with any excess material trimmed away.

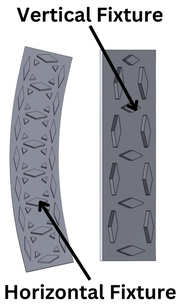


Magnetization

Following the fabrication of the unit cells, they were positioned in a 3D-printed fixture that arranged them in a diamond-like pattern. These cells were placed inside a 1.5T magnetic impulse field to establish the predetermined magnetization directions.



Assembly



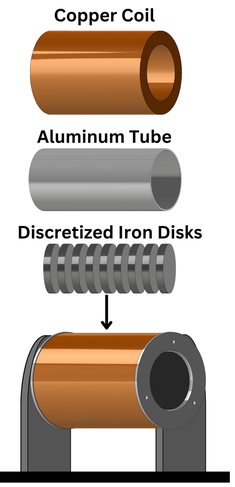
Once magnetized, the array was assembled by connecting the hinges of the unit cells with silicone epoxy. The cells were first connected vertically and then horizontally. Using the labeling system in Figure 1, all UC1 cells were assembled vertically using a custom 3D-printed fixture. Subsequently, UC1-UC5 were connected horizontally using horizontal fixtures.

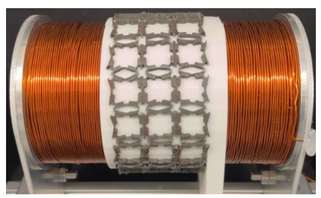
Uniform cells with precise connection points were used throughout the cylindrical configuration. Connections were made at angles to best fit the curved 3D-printed shapes for the concave and convex configurations,

Discrete Iron Core Electromagnetic Coil

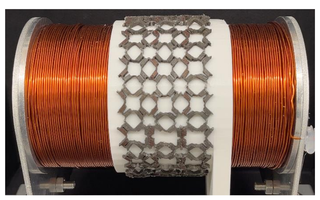
The hollow circular structure, composed of 16x3 cells, was placed around a magnetic coil with an iron core. The unit cells contained 35% magnetic particles and were actuated using a current range of 10-22.5A. Despite the iron core coil typically being stronger than an ironless coil, it was the smallest in this setup. The magnetic particle percentage was increased to balance the magnetic field interactions and achieve even actuation, similar to the actuation in the convex, concave, and cylindrical substrates.

The iron coil was made up of three important components. First, the copper coil was evenly wound against the aluminum tube, and then discretized iron disks were evenly distributed along the inside core of the coil. These components combined created an even magnetic field.





Deployed

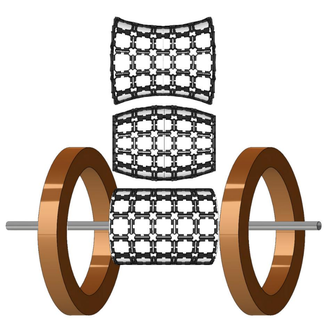


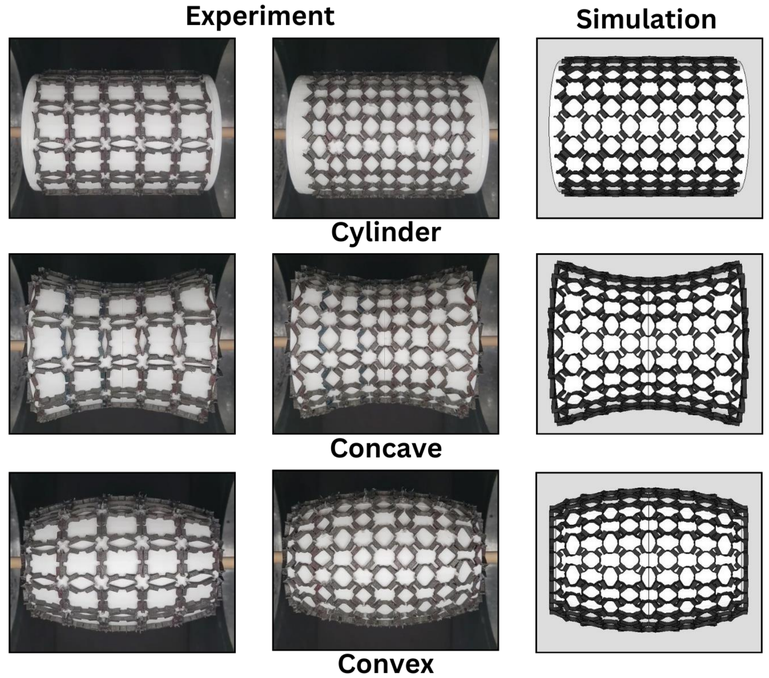
Folded

At 22.5A, peak actuation was achieved, showing the effectiveness of the magnetic field in inducing deformation. The iron core field strength impacted the responsiveness of the magneto-metamaterial. The increased percentage of magnetic particles allowed for more pronounced deformation, highlighting the importance of particle concentration in achieving the desired actuation. These results show that fine-tuning the magnetic particle concentration and applied current, combined with the iron core, can optimize magneto-metamaterial performance, providing insights into the design of future systems.

Cylinder, Concave, Convex Actuation

The second actuation involved metamaterials in concave, convex, and cylindrical shapes, each with a 12x5 grid of cells containing 25% magnetic powder. The same magnetic field was applied to each test. Two magnetic coils were placed on either side of the substrate to actuate these structures.





The same manufacturing methods and actuation procedures were used for the convex, concave, and cylindrical structures. The unit cells were shaped and angled differently to match each substrate's curvature, allowing the metamaterials to better conform to the 3D shapes. Results demonstrated the feasibility of creating functional metamaterials that adapt to curved edges. This finding highlights the potential of metamaterials to achieve precise deformation patterns and conform to various geometries.

Simulation

The cell arrays and substrates underwent a two-step simulation and experimental testing process to ensure accuracy. Simulations using Abaqus involved detailed meshing of each cell partition. The cell arrays were then experimentally tested with the coils. The experimental results consistently matched the simulation predictions, confirming the precision of our models.

Future Studies & Conclusion

The study demonstrated the potential of magnetically actuated metamaterials across various 3D substrates, influenced by the unit cell design. Results showed that optimizing magnetic particle concentration and applied current is crucial for achieving desired deformation patterns. Future research should focus on testing the efficiency of magneto-metamaterials to cloak wave transmission or steer acoustic waves.[1] Exploring new design and manufacturing approaches for unit cell configurations and overall structure can enhance metamaterial performance. The field is still developing, with opportunities for further investigation into manufacturing techniques, design methodologies, and practical applications.

Acknowledgments

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References

[1] Lu, L., Sim, J., & Zhao, R. R. (2024). Mechanics of hard-magnetic soft materials: A Review. *Mechanics of Materials*, *189*, 104874 https://doi.org/10.1016/j.mechmat.2023.104874

[2] Sim, J., Wu, S., Dai, J., Zhao, R. R., Magneto-Mechanical Bilayer Metamaterial with Global Area-Preserving Density Tunability for Acoustic Wave Regulation. *Adv. Mater.* 2023, 2303541. https://doi.org/10.1002/adma.202303541

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